METHODOLOGY PASSPORT CARD

Revised Research Design

Research topic: Trust in AI-Generated Medical Recommendations Among Patients: A Comparison of Three Age Groups

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1. Research Type

Quantitative, observational, cross-sectional survey of patients aged 18+. Groups: 18–30, 31–50 and 51+. The dataset in this package is simulated; real data collection is planned.

2. Research Questions and Hypotheses

RQ1: Does mean trust differ across the three patient age groups? **H0:** All three population means are equal. **H1:** At least one differs.

RQ2: Are AI knowledge, perceived accuracy, explanation clarity and doctor supervision associated with trust, adjusting for age group? **H0:** All adjusted associations are zero. **H1:** At least one is nonzero.

RQ3: Is prior AI use associated with higher trust after accounting for age group? **H0:** No association. **H1:** Positive association.

3. Variable Matrix

Role	Operational definition
Independent	Age group: 18–30 / 31–50 / 51+
Dependent	Trust in AI medical recommendations (1–5)
Secondary predictors	Prior AI use (yes/no); knowledge, accuracy, clarity, doctor supervision (1–5)
Planned controls	Education, healthcare experience, digital literacy, medical-app experience

4. Metrics and Guardrails

Primary: age-group means and group differences with 95% CIs. Secondary: median, SD, trust by prior AI use and perception ratings. Guardrails: valid completion ≥90%; invalid/duplicate ≤10%; exclude identifiable or sensitive diagnoses.

5. Baseline and Planned Analysis

Age group 18–30 serves as the **descriptive reference group** for age contrasts, not a randomized control. With real data, consider ANOVA/Kruskal–Wallis and appropriate adjusted models; plan ≥30 valid participants per group (not a formal power calculation).

6. Ethics and Limitations

Obtain voluntary informed consent and protect privacy. The 45 simulated records (15/group) test the analysis pipeline only. They cannot validate any hypothesis or support conclusions about real patients.